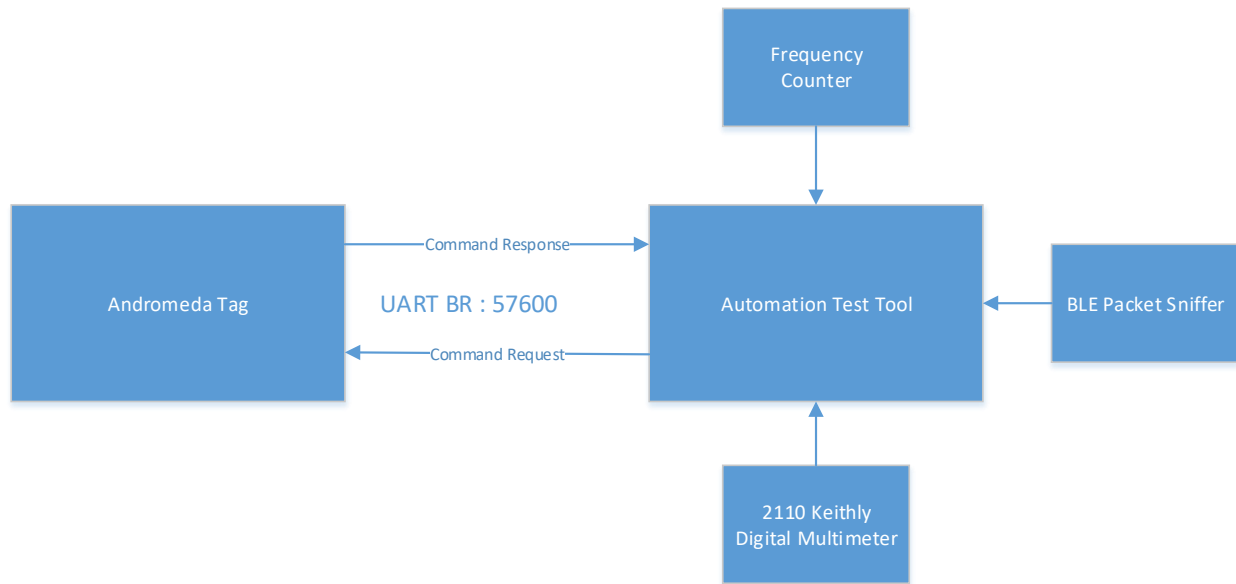


Automation Test:



Hardware Requirement:

- Frequency Counter
- 2110 Keithly Digital Multimeter
- BLE Packet Sniffer (CC2540 Dongle)
- Monitor
- Exciter
- USB to Serial Dongle.

Software Requirement:

- Mini Circuit Frequency counter (Frequency Counter)
- TI Packet Sniffer (CC2540 Dongle)
- IviSharedComponents (2110 Keithly Digital Multimeter)
- Ke2100 (2110 Keithly Digital Multimeter)
- NIVISA1550full (2110 Keithly Digital Multimeter)

Note: Comm port should be less than comm 10.

Automation Test :

- Tag enter into ATE mode whenever Tag has been reset
- In ATE Mode, Tag is waiting for 2 sec to receive command from ATE Test Tool
- If Tag didn't receive command from ATE Test Tool within 2 sec then Tag moves normal RTLS Process.
- ATE Test Tool sent Enter ATE mode request to Tag up to 20 sec. If Tool receives response from Tag then Tool stop sending the request.
- Tag enters into Automation Test mode when Tag receive "Enter ATE mode" request from tool.
- Red Led blinks (On time: 20 ms , offtime : 250 ms) upto 10 sec. This indicates that tag was in ATE Mode and waiting for command request.
- Tag switch to each Test based on the command received from Test tool
- Test tool read data from Frequency counter, Multimeter and BLE Packet sniffer based on the test and show the result.
- we can configure the Criteria in Settings.ini file.
- We can Enable or Disable the each test based on INI Configuration

RF Transmission Test:

- **Tag** : RF CW Tx = 914 MHZ, Tx Power = 10 dbm, This will be configure in settings.ini file
- **Tool**: Frequency counter used to read the Tag transmitted frequency. Test will pass When the frequency within the range configure in ini file
- **Allowed offset**: Difference b/w the Allowed Frequency and Actual tag transmitted frequency(ie: frequency measure in frequency counter) is called as Allowed offset. Test will Pass when the Offset is less then allowed offset.
- **Timeout** : configurable in ini file Default = 1000 ms, Tag Transmit CW Tx upto 1000 ms
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : TX_Frequency, Tx_Power, Allowed Frequency,Allowed offset,Timeout, Retry, Enable/Disable the Test
- **Enable calibration**: we can enable or disable the RF calibration. If it is enabled then Tag will calibrated only when test was failed.

```

[RF_TRANSMISSION_TEST]

TX_POWER = Value (1 = -10dbm, 2 = 0dbm, 3 = 1dbm, 4 = 2dbm, 5 = 3dbm, 6 = 4dbm,
                  7 = 5dbm, 8 = 6dbm, 9 = 7dbm, 10 = 8dbm, 11 = 9dbm, 12 = 10dbm,
                  13 = 11dbm, 14 = 12dbm, 15 = 14dbm )

[RF_TRANSMISSION_TEST]
NAME = RF Transmission Test
ALLOWD_FREQ = 914.000
ALLOWED_OFFSET = 10
TX_FREQ = 1
TX_POWER = 12
TIMEOUT = 1000
RETRY = 3
ENABLE_TEST = 1
ENABLE_CALIBRATION = 0

```

RF Sensitivity Test:

- **Tag** : RF packet Rx = 914 MHZ, This will be configure in settings.ini file
- **Tool**: Test Monitor Transmit RF Packet in 500 ms. Tag receives this packet sent received signal strength RSSI to Tool. Test will pass when the RSSI is within range
- **Timeout** : configurable in ini file Default = 1000 ms, Tag Packet Rx upto 1000 ms
- **Retry**: configurable in ini file Default = 1
- **INI Configuration** :RX_Freq, Low/high Rssi_Range ,Timeout, Retry, Enable/Disable the Test

```

[RF_SENSITIVITY_TEST]
[RF_SENSITIVITY_TEST]
NAME = RF Sensitivity Test
RX_FREQ = 1
LOW_RSSI_RANGE = -87
HIGH_RSSI_RANGE = -70
TIMEOUT = 1000
RETRY = 3
ENABLE_TEST = 1

```

Battery Voltage Test:

- **Tag** : Battery Monitor in Tag used to measure battery voltage
- **Tool**: Tag sent measured Battery voltage to Tool. Test will pass when the battery voltage is within range
- **Timeout** : configurable in ini file Default = 1000 ms, Tag measure Battery upto 1000 ms with 250 ms gap
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Battery Voltage Range, Timeout, Retry, Enable/Disable the Test

```

: [VOLTAGE_TEST]
:
:
:
:
: [VOLTAGE_TEST]
: NAME = Battery Voltage Test
: MIN_VOLTAGE = 3.15
: MAX_VOLTAGE = 3.40
: TIMEOUT = 1000
: RETRY = 3
: ENABLE_TEST = 1

```

BLE Tx Test:

- **Tag** : BLE Tx in channel : 37, 38, 39, Tx Power = +5 dbm
- **Tool**: Tool receives BLE Packet from BLE Packet sniffer. Test will Pass only when no. of received packets was higher than the count value.
- **Timeout** : configurable in ini file Default = 3000 ms, Tag Transmit BLE Packet upto 3000 ms with 50 ms interval
- **Retry**: configurable in ini file Default = 1,
- **INI Configuration** : Tx Power, Count,Port,Timeout, Retry, Enable/Disable the Test

```
;;
; [BLE_TX_TEST]
; TX_POWER = Value( 0 = -20 dbm, 1 = - 18 dbm, 2 = -15 dbm, 3 = -12 dbm, 4 = -10 dbm,
;                  5 = -9 dbm, 6 = -6 dbm, 7 = -5 dbm, 8 = -3 dbm,
;                  9 = 0, 10 = 1, 11 = 2, 12 = 3, 13 = 4, 14 = 5 )
;
;
;
;;
[BLE_TX_TEST]
NAME = BLE Tx Test|
TIMEOUT = 3000
RETRY = 3
PORT = 7000
ENABLE_TEST = 1
TX_POWER = 14
COUNT = 2
```

Accelerometer Test:

- **Tag** : Tag read the accelerometer value Range, Resolution, X, Y & Z. Tag Sent this value to ATE Tool
- **Tool**: ATE Tool receive this value and calculation gravity. If the gravity value within the configured XY&Z range then the test will be pass.
- **Timeout** : configurable in ini file Default = 1000 ms, Motion the Tag with in 1000 ms.
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Timeout, Retry, Enable/Disable the Test

```
;;
; [ACCELEROMETER_TEST]
;
;
;
;;
[ACCELEROMETER_TEST]
NAME = Accelerometer Test
TIMEOUT = 1000
RETRY = 3
ENABLE_TEST = 1

X_G_RANGE_FROM = -10.0
X_G_RANGE_TO = 10.0
Y_G_RANGE_FROM = 10.0
Y_G_RANGE_TO = 10.0
Z_G_RANGE_FROM = -10.0
Z_G_RANGE_TO = 10.0
```

Buzzer Test:

- **Tag** : Tag Buzzer Buzz up to timeout configure in ini file
- **Tool**: Tool measure the buzzer current from Keithly Digital Multimeter. Test will Pass when current is within the range
- **Timeout** : configurable in ini file Default = 1000 ms, Buzzer on up to 1000 ms.
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Current Range, Timeout, Retry, Buffer_Freq,Enable/Disable the Test

```
;; ;;;
; [BUZZER_TEST]
;
;
;
; ;;;
[BUZZER_TEST]
NAME = Buzzer Test
MIN_CURRENT = 0.80 ;;mA
MAX_CURRENT = 1.05 ;;mA
TIMEOUT = 1000
RETRY = 3
ENABLE_TEST = 1
BUZZER_FREQ = 4.5
```

Flash Memory Test:

- **Tag** : Tag write & read 256 byte in external flash memory
- **Tool**: Tool receives the status whether write & read bytes are same or not. Test will pass when Both read and write are same/matched
- **Timeout** : configurable in ini file Default = 500 ms, Tag write & read external flash up to 500 ms.
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Timeout, Retry, Enable/Disable the Test

```
;; ;;;
; [FLASH_MEMORY_TEST]
;
;
;
; ;;;
[FLASH_MEMORY_TEST]
NAME = Flash Memory Test
TIMEOUT = 500
RETRY = 3
ENABLE_TEST = 1
```

LF Calibration Test:

- **Tag** : Tag auto tune LF Capacitor
- **Tool**: Tool receives the tuned status & capacitor value. Test will pass when capacitor value in the range
- **Timeout**: configurable in ini file Default = 2000 ms, Tag write & read external flash up to 2200ms.
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Timeout, Retry, Enable/Disable the Test

```

[LF_CALIBRATION_TEST]
|
[LF_CALIBRATION_TEST]
NAME = LF calibration Test
TIMEOUT= 2000
RETRY= 1
ENABLE_TEST= 0

```

LF RX Channe1 Test:

- **Tag** : Tag receives LF Id from Exciter via channel 1
- **Tool**: Tool receives the received LF Id from Tag. Test will pass when received LF Id & configure LF Id are same.
- **Timeout** : configurable in ini file Default = 3000 ms, Tag receives LF Id up to 3000 ms.
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : LF Id, Timeout, Retry, Enable/Disable the Test

```

[LF_RX_CHANNLE1_TEST]

[LF_RX_CHANNLE1_TEST]
NAME = LF RX Channel 1 Test
LF_ID = 2272
TIMEOUT = 3000
RETRY = 1
ENABLE_TEST = 1

```

LF RX Channe3 Test:

- **Tag** : Tag receives LF Id from Exciter via channel 3
- **Tool**: Tool receives the received LF Id from Tag. Test will pass when received LF Id & configure LF Id are same.
- **Timeout** : configurable in ini file Default = 3000 ms, Tag receives LF Id up to 3000 ms.
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : LF Id, Timeout, Retry, Enable/Disable the Test

```
;; ;;;
; [LF_RX_CHANNLE3_TEST]
;
;
;
;; ;;;
[LF_RX_CHANNLE3_TEST]
NAME = LF RX Channe] 3 Test
LF_ID = 2272
TIMEOUT = 3000
RETRY = 1
ENABLE_TEST = 1
```

IR RX A Test:

- **Tag** : Tag receives IR Id from Monitor
- **Tool**: Tool receives the received IR Id from Tag. Test will pass when received IR Id & configure LF Id are same.
- **Timeout** : configurable in ini file Default = 6000 ms, Tag receives LF Id up to 6000 ms.
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : IR_Id, Timeout, Retry, Enable/Disable the Test

```
;; ;;;
; [IR_RXA_TEST]
;
;
;
;; ;;;
[IR_RXA_TEST]
NAME = IR RX A Test
IR_ID = 2232
TIMEOUT = 6000|
RETRY = 1
ENABLE_TEST = 1
```

IR RX B Test:

- **Tag** : Tag receives IR Id from Monitor
- **Tool**: Tool receives the received IR Id from Tag. Test will pass when received IR Id & configure LF Id are same.
- **Timeout** : configurable in ini file Default = 6000 ms, Tag receives LF Id up to 6000 ms.
- **Retry**: configurable in ini file Default = 1.0
- **INI Configuration** : IR_Id, Timeout, Retry, Enable/Disable the Test

```
;; ;;;
; [IR_RXB_TEST]
;
;
;; ;;;
[IR_RXB_TEST]
NAME = IR RX B Test
IR_ID = 2232
TIMEOUT = 6000
RETRY = 1
ENABLE_TEST = 1
```

Led 1 Test:

- **Tag** : Tag turn on led 1.
- **Tool**: Tool measure the Led 1 current from Keithly Digital Multimeter. Test will Pass when current is within the range
- **Timeout** : configurable in ini file Default = 500 ms, Tag Turn on Led 1 up to 500 ms.
- **Retry**: configurable in ini file Default = 1.
- **Led_out_current**: we can control the led on current .for eg: Led_outIcurrent = 7 then 1 mA current when Led was turned on
- **INI Configuration** : Current Range, Timeout, Retry, Enable/Disable the Test

```
;; ;;;
; [LED1_TEST]
;
; 7 - 1 mA
; 79 - 10 mA
;; ;;;
[LED1_TEST]
NAME = Led 1 Test
MIN_CURRENT = 1.00 ;;mA
MAX_CURRENT = 1.75 ;;mA
TIMEOUT = 500
RETRY = 3
ENABLE_TEST = 1
LED_OUT_CURRENT = 7
```


Led 2 Test:

- **Tag** : Tag turn on led 2
- **Tool**: Tool measure the Led 2 current from Keithly Digital Multimeter. Test will Pass when current is within the range
- **Timeout** : configurable in ini file Default = 500 ms, Tag Turn on Led 2 up to 500 ms.
- **Retry**: configurable in ini file Default = 1.
- **Led_out_current**: we can control the led on current .for eg: Led_outcurrent = 7 then 1 mA current when Led was turned on
- **INI Configuration** : Current Range, Timeout, Retry, Enable/Disable the Test

```
[LED2_TEST]

[LED2_TEST]
NAME = Led 2 Test
MIN_CURRENT = 1.00 ;;mA
MAX_CURRENT = 1.75 ;;mA
TIMEOUT = 500
RETRY = 3
ENABLE_TEST = 1
LED_OUT_CURRENT = 7
```

Led 3 Test:

- **Tag** : Tag turn on led 3
- **Tool**: Tool measure the Led 3 current from Keithly Digital Multimeter. Test will Pass when current is within the range
- **Timeout** : configurable in ini file Default = 500 ms, Tag Turn on Led 3 up to 500 ms.
- **Retry**: configurable in ini file Default = 1.
- **Led_out_current**: we can control the led on current .for eg: Led_outcurrent = 7 then 1 mA current when Led was turned on .
- **INI Configuration** : Current Range, Timeout, Retry, Enable/Disable the Test

```

[LED3_TEST]

[LED3_TEST]
NAME = Led 3 Test
MIN_CURRENT = 1.00 ;;mA
MAX_CURRENT = 1.75 ;;mA
TIMEOUT = 500
RETRY = 3
ENABLE_TEST = 1
LED_OUT_CURRENT = 7

```

Led 4 Test:

- **Tag** : Tag turn on led 4
- **Tool**: Tool measure the Led 4 current from Keithly Digital Multimeter. Test will Pass when current is within the range
- **Timeout** : configurable in ini file Default = 500 ms, Tag Turn on Led 4 up to 500 ms.
- **Retry**: configurable in ini file Default = 1.
- **Led_out_current**: we can control the led on current .for eg: Led_outlcurrent = 7 then 1 mA current when Led was turned on .
- **INI Configuration** : Current Range, Timeout, Retry, Enable/Disable the Test

```
;;::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::;;
; [LED4_TEST]
;
;
;
;
;;::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::;;
[LED4_TEST]
MIN_CURRENT = 1.00 ;;mA
MAX_CURRENT = 1.75 ;;mA
TIMEOUT = 500
RETRY = 3
ENABLE_TEST = 1
LED_OUT_CURRENT = 7
```

RED Led Test:

- **Tag** : Tag turn on RED Led
- **Tool**: Tool measure the RED Led current from Keithly Digital Multimeter. Test will Pass when current is within the range
- **Timeout** : configurable in ini file Default = 1000 ms, Tag Turn on RED up to 1000 ms.
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Current Range, Timeout, Retry, Enable/Disable the Test

```
;;::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::;;
; [RED_LED_TEST]
;
;
;
;
;;::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::;;
[RED_LED_TEST]
NAME = Red Led Test
MIN_CURRENT = 2.00 ;;mA
MAX_CURRENT = 2.90 ;;mA
TIMEOUT = 1000
RETRY = 3
ENABLE_TEST = 1
```

BUTTON 1 Test:

- **Tag** : Button 1 was pressed before this Button 1 Test.
- **Tool**: Tool receives pressed buttons status from Tag. Tool validate that Button 1 was pressed before Button 2 and Button 3(Ie: Tool validate key order 1,2&3) and then Test will be pass
- **Timeout** : configurable in ini file Default = 5000 ms, Press Button manually within 5000 ms
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Key Value, Timeout, Retry, Enable/Disable the Test

```
;; .....  
; [BUTTON_1_TEST]  
;  
;  
;  
;; .....  
[BUTTON_1_TEST]  
NAME = Button 1 Test  
KEY_VALUE = 1  
TIMEOUT = 5000  
RETRY = 3  
ENABLE_TEST = 1
```

BUTTON 2 Test:

- **Tag** : : Button 2 was pressed before this Button 2 Test.
- **Tool**: Tool receives pressed buttons status from Tag. Tool validate that Button 2 was pressed after Button 1 and before Button 3(Ie: Tool validate key order 1,2&3) and then Test will be pass
- **Timeout** : configurable in ini file Default = 5000 ms, Press Button manually within 5000 ms
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Key Value, Timeout, Retry, Enable/Disable the Test

```
;; .....  
; [BUTTON_2_TEST]  
;  
;  
; |  
;; .....  
[BUTTON_2_TEST]  
NAME = Button 2 Test  
KEY_VALUE = 2  
TIMEOUT = 5000  
RETRY = 3  
ENABLE_TEST = 1
```

BUTTON 3 Test:

- **Tag** : Button 3 was pressed before this Button 3 Test.
- **Tool**: Tool receives pressed buttons status from Tag. Tool validate that Button 3 was pressed after Button 1 and Button 2(ie: Tool validate key order 1,2&3) and then Test will be pass
- **Timeout** : configurable in ini file Default = 5000 ms, Press Button manually within 5000 ms
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Key Value, Timeout, Retry, Enable/Disable the Test

```

[BUTTON_3_TEST]

[BUTTON_3_TEST]|
NAME = Button 3 Test
KEY_VALUE = 3
TIMEOUT = 5000
RETRY = 3
ENABLE_TEST = 1

```

Motion Test:

- **Tag** : Tag wait for Manual Motion
- **Tool**: if Tag get motion then sent the motion information to Tool. Test will Pass when Tag was manual motion
- **Timeout** : configurable in ini file Default = 3000 ms, Motion the Tag with in 3000 ms.
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Timeout, Retry, Enable/Disable the Test

```
[MOTION_TEST]
NAME = Motion Test
TIMEOUT = 3000
RETRY = 3
ENABLE_TEST = 0
```

Proximity Test:

- **Tag** : Tag Measure the Proximity value
- **Tool**: Tool receives measured proximity value from Tag. Tag will Pass when the proximity value within the range
- **Timeout** : configurable in ini file Default = 1000 ms, Motion the Tag with in 1000 ms.
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Proximity Range, Timeout, Retry, Enable/Disable the Test

[illegible]

Sleep Current Test:

- **Tag** : Tag enter into sleep current mode for 1.5 ms
- **Tool**: Tool measure the sleep current from Keithly Digital Multimeter. Test will Pass when current is within the range
- **Timeout** : configurable in ini file Default = 1000 ms, Tag was in Sleep current upto 1000 ms
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Current Range, Timeout, Retry, Enable/Disable the Test

```

; [SLEEP_CURRENT_TEST]
;
;
;
; [SLEEP_CURRENT_TEST]
NAME = Sleep Current Test
MIN_CURRENT = 6 ;;uA
MAX_CURRENT = 10 ;;uA
TIMEOUT = 1000
RETRY = 3
ENABLE_TEST = 1

```

Get Factory Param:

- **Tag** : Tag Sent Factory settings parameter information to Tool.
- **Tool**: when Test will always pass Tool will Receive and display the factory Param.
- **Timeout** : configurable in ini file Default = 1000 ms
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Timeout, Retry, Enable/Disable the Test

```
.....  
; [FACTORY_PARAM]  
;  
;  
;  
.....  
[FACTORY_PARAM]  
NAME = Get Factory Param  
TIMEOUT = 1000  
RETRY = 3  
ENABLE_TEST = 1
```

32 kHz Clock Test:

- **Tag** : 32 KHZ was Generate in Pin TDO(DIO 16).
- **Tool**: Display only test will be pass. There is no Response from tag.
- **Timeout** : configurable in ini file Default = 5000 ms
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Timeout, Retry, Enable/Disable the Test

```
.....  
; [32KHZ_CLOCK_TEST]  
;  
;  
;  
.....  
[32KHZ_CLOCK_TEST]  
NAME = 32 kHz Clock Test  
TIMEOUT = 5000  
RETRY = 3  
ENABLE_TEST = 1
```

250 ms Clock Test:

- **Tag** : 250 ms was Generate in Pin TDI(DIO 17).
- **Tool**: Display only test will be pass. There is no Response from tag.
- **Timeout** : Configurable in ini file Default = 5000 ms
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Timeout, Retry, Enable/Disable the Test, Clock count

```
;; ;;;
; [250MS_CLOCK_TEST]
;
;
;
;; ;;;
[250MS_CLOCK_TEST]
NAME = 250 ms Clock Test
TIMEOUT = 5000
RETRY = 1
ENABLE_TEST = 1
CLOCK_COUNT = 4
```

Factory Sleep:

- **Tag** : whenever tag will receive the factory sleep command from Tool then tag will go the factory sleep mode.
- **Tool**: If all test was passed in tool then only tool send the factory sleep command.
- **Timeout** : configurable in ini file Default = 500 ms
- **Retry**: configurable in ini file Default = 1.
- **INI Configuration** : Timeout, Retry, Enable/Disable the Test

```
;; ;;;
; [FACTORY_SLEEP_TEST]
;
; |
;
;
;; ;;;
[FACTORY_SLEEP_TEST]
NAME = Factory Sleep
TIMEOUT = 500
RETRY = 1
ENABLE_TEST = 1
```